CH4 Distribution Functions and Discrete Random Variables

4.1 Random variables

Def A real-valued function X: S→R is called a random variable of the experiment if, for each interval I ⊆ R, $\{s : X(s) \in I\}$ is an event.

In probability, the set $\{s : X(s) \in I\}$ is often abbreviated as $\{X \in I\}$ or simply as $X \in I$

- **Ex** Three balls are selected randomly without replacement from a box containing 20 balls number 1 through 20. What is the probability that the largest number of the drawn balls is as large as or larger than 17?

Sol: Let X denote the largest drawn number

Sol. (a)

$$P(X=k)=\frac{\binom{k-1}{2}}{\binom{20}{3}},\ k=3,4,\ldots,20 \Rightarrow P(X\geq 17)=\sum_{k=17}^{20}P(X=k)$$

Sol. (b)

$$P(X\geq 17)=1-P(X\leq 16)=1-\frac{\binom{16}{3}}{\binom{20}{3}},$$

4.2 Distribution functions

CDF: (cumulative) distribution function

$$F_x(t) = F(t) = P(X \leq t)$$

1. F is nondecreasing; that is, if t<u, then F(t)≤F(u).
2. $\lim_{t\to\infty} F(t) = 1 \qquad \lim_{t\to-\infty} F(t) = 0$
3. F is right continuous; that is, for every t in R, $F(t^+) = F(t)$

Event concerning X	Probability in terms of F	Event concerning X	Probability in terms of F
$X \leq a$	$F(a)$	$a < X \leq b$	$F(b) - F(a)$
$X > a$	$1 - F(a)$	$a < X < b$	$F(b^-) - F(a)$
$X < a$	$F(a^-)$	$a \leq X \leq b$	$F(b) - F(a^-)$
$X \geq a$	$1 - F(a^-)$	$a \leq X < b$	$F(b^-) - F(a^-)$
$X = a$	$F(a) - F(a^-)$		

4.3 Discrete random variables

Discrete (離散的)

A set of values that is countable is called discrete

Countable: 有空隙在每個數值之間，才可以數

Examples of set measure

- Finite: {0, 1, 2}
- Finite: {1, 2, 3, 4, … }
- Infinite: {x : x > 0} (You can not find what is the next of zero, 0.1 0.01 ?)

P.F.M (Probability mass function)

(a) $p(x) = 0 \text{ if } x \notin \{x_1, x_2, x_3, ...\}$

(b) $p(x_i) = P(X = x_i) \text{ and hence } p(x_i) \geq 0$

(c) $\sum_{i=1}^{\infty} p(x_i) = 1$

4.4 Expectations of discrete random variables

Expected value

$$E(X) = \Sigma_{x \in A} x p(x)$$

Called: mean, expectation of X

Denoted: EX, μ_x, μ

Ex 4.19 The tanks of a country's army are numbered 1 to N. In a war this country loses n random tanks to the enemy, who discovers that the captured tanks are numbered. If X1 , X2, …, Xn are the numbers of the captured tanks, what is E(max Xi)? How can the enemy use E(max Xi) to find an estimate of N, the total number of this country's tanks?

Let $Y = \max X_i$ for $k \in n, n+1, n+2, ..., N$

$$P(Y = k) = \frac{C_{n-1}^{k-1}}{C_n^N}$$

$$E(Y) = \Sigma_{k=n}^{N} k \times P(Y = k)$$

$$= \sum_{k=n}^{N} k \times \frac{(k-1)!}{(n-1)! \cdot (k-n)!} \times \frac{1}{C_n^N}$$

$$= n \cdot \frac{1}{C_n^N} \cdot \sum_{k=n}^{N} \frac{k!}{n! \cdot (k-n)!}$$

$$= n \cdot \frac{1}{C_n^N} \cdot \sum_{k=n}^{N} C_n^k$$

$$= n \cdot \frac{1}{C_n^N} \cdot (C_n^n + C_n^{n+1} + C_n^{n+2} + \cdots + C_n^N)$$

$$= n \cdot \frac{1}{C_n^N} \cdot C_{n+1}^{N+1} \#$$

$$N = \frac{1}{n} \cdot C_n^N \cdot \frac{1}{C_{n+1}^{N+1}} \cdot E(Y) - 1$$

$$= \frac{n+1}{n} E(Y) - 1$$

$N!$

E(Y)代入觀察到的最大編號

n 代入樣本數量

$$= C_n^n + C_n^{n+1} + \cdots + C_n^N$$

$$= C_{n+1}^{n+1} + C_n^{n+1}$$

$$= C_{n+1}^{n+2} + C_n^{n+2}$$

$$= C_n^{n+3} + C_n^{n+3}$$

$$\vdots$$

$$= C_{n+1}^{N+1}$$

Theorem 4.1

If X is a random variable and P(X=c)=1, then E(X) = c

Theorem 4.2

$$E[g(x)] = \Sigma_{x \in A}\ g(x) p(x)$$

Coro

E(X) obeys the linear rules: $E(aX + b) = aE(X) + b$

4.5 Variances and moments of discrete random variables

Definition: Variance of X

$$Var(X) = \sigma_x^2 = E[(X - E(X))^2] = E[(E - \mu)^2] = \Sigma_x (x - \mu)^2 p(x)$$

Standard deviation of X: $\sigma_x = \sqrt{Var(x)}$

Theorem 4.3

$$Var(X) = E(X^2) - (E(X))^2$$

Application: $(E(X))^2 \leq E(X^2)$

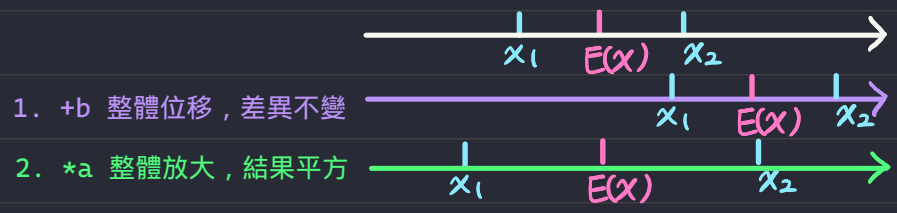
Theorem 4.4

Var(X)=0 if and only if X is constant with probability 1

Theorem 4.5

$$Var(aX + b) = a^2 Var(X)$$

→ Var 在計算實際情與平均值的差異（距離）



Definition

Let w be a given point. X is more concentrated about w than is Y. For all t > 0

$$P(|Y - w| \leq t) \leq P(|X - w| \leq t)$$

Theorem 4.6

Suppose that $E(X) = E(Y) = a$

If X is more concentrated about a than is Y, then $Var(X) \leq Var(Y)$

Definition

$\mathbf{E[g(X)]}$	**Definition**
$E(X^n)$	The nth moment of X
$E(\|X\|^r)$	The rth absolute moment of X
$E(X - c)$	The 1st moment of X about c
$E[(X - c)^n]$	The nth moment of X about c
$E[(X - E(X))^n]$	The nth central moment of X

4.6 Standardized random variables

Definition

The standardized X denote $X^* = (X - \mu)/\sigma$ where μ is the mean σ is 標準差

$E(X^*) = E\left(\frac{1}{\sigma}X - \frac{\mu}{\sigma}\right) = \frac{1}{\sigma}E(X) - \frac{\mu}{\sigma} = 0$

$Var(X^*) = Var\left(\frac{1}{\sigma}X - \frac{\mu}{\sigma}\right) = \frac{1}{\sigma^2}Var(X) = 1$